Experiment 8:

```
Mar 31 14:47
rvu@rvu-OptiPlex-SFF-7020: ~/shourrya_0100/operating_systems
  GNU nano 7.2                                    ev.sh
#!/bin/bash

display_env()
{
        echo "Current Environment Variables"
        printenv
}

set_env()
{
read -p "Enter variable names: " name
read -p "Enter Variable Values: " value

export $name=$value
echo "Variable $name set successfully"
}

show_env()
{
echo "Environment variables using env command: "
env
}

while true
do
 echo "-----Environment Variables Manager------"
 echo "1.Display Variables(printenv)"
 echo "2. Set New Variables "
 echo "3.Display variables(env)"
 echo "4.Exit"

read -p "Enter choice: " ch

^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute   ^C Location   M-U Undo      M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^Q Where Was
```

```
Mar 31 14:48
rvu@rvu-OptiPlex-SFF-7020: ~/shourrya_0100/operating_systems
  GNU nano 7.2                                    ev.sh
read -p "Enter variable names: " name
read -p "Enter Variable Values: " value

export $name=$value
echo "Variable $name set successfully"
}

show_env()
{
echo "Environment variables using env command: "
env
}

while true
do
 echo "-----Environment Variables Manager------"
 echo "1.Display Variables(printenv)"
 echo "2. Set New Variables "
 echo "3.Display variables(env)"
 echo "4.Exit"

read -p "Enter choice: " ch

case $ch in
        1) display_env ;;
        2) set_env ;;
        3) show_env ;;
        4) exit ;;
        *) echo &quot;Invalid choice&quot; ;;
esac

done

^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute   ^C Location   M-U Undo      M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^Q Where Was
```

```
Mar 31  14:51
rvu@rvu-OptiPlex-SFF-7020: ~/shourrya_0100/operating_systems
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$ ls
 abc.txt              child_pro.sh   hi.txt   lab31.sh         owner.sh   perm3.sh             process_creation_1RUA25BCA0100.pdf
 background_task.sh   cp.sh          IO.sh    lab3-2.sh        perm1.sh   perm.sh              process.sh
 child_1.sh           ev.sh          l3.sh    lu6999oyuj.tmp   perm2.sh   process_control.sh  'Screenshot from 2026-02-17 14-12-43-2.pdf'
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$ nano ev.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$ chmod +x ev.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$ ./ev.sh
-----Environment Variables Manager------
1.Display Variables(printenv)
2. Set New Variables
3.Display variables(env)
4.Exit
Enter choice: 1
Current Environment Variables
SHELL=/bin/bash
SESSION_MANAGER=local/rvu-OptiPlex-SFF-7020:@/tmp/.ICE-unix/2389,unix/rvu-OptiPlex-SFF-7020:/tmp/.ICE-unix/2389
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
CONDA_EXE=/home/rvu/anaconda3/bin/conda
_CE_M=
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
MEMORY_PRESSURE_WRITE=c29tZSAyMDAwMDAgMjAwMDAwMAA=
XMODIFIERS=@im=ibus
DESKTOP_SESSION=ubuntu
GTK_MODULES=gail:atk-bridge
PWD=/home/rvu/shourrya_0100/operating_systems
GSETTINGS_SCHEMA_DIR=/home/rvu/anaconda3/share/glib-2.0/schemas
XDG_SESSION_DESKTOP=ubuntu
LOGNAME=rvu
XDG_SESSION_TYPE=wayland
CONDA_PREFIX=/home/rvu/anaconda3
SYSTEMD_EXEC_PID=2437
GSETTINGS_SCHEMA_DIR_CONDA_BACKUP=
```

```
Mar 31  14:51
rvu@rvu-OptiPlex-SFF-7020: ~/shourrya_0100/operating_systems
/bin:/snap/bin
GDMSESSION=ubuntu
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
OLDPWD=/home/rvu/shourrya_0100
_=/usr/bin/printenv
-----Environment Variables Manager------
1.Display Variables(printenv)
2. Set New Variables
3.Display variables(env)
4.Exit
Enter choice: 2
Enter variable names: hello
Enter Variable Values: 100
Variable hello set successfully
-----Environment Variables Manager------
1.Display Variables(printenv)
2. Set New Variables
3.Display variables(env)
4.Exit
Enter choice: 3
Environment variables using env command:
SHELL=/bin/bash
SESSION_MANAGER=local/rvu-OptiPlex-SFF-7020:@/tmp/.ICE-unix/2389,unix/rvu-OptiPlex-SFF-7020:/tmp/.ICE-unix/2389
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
CONDA_EXE=/home/rvu/anaconda3/bin/conda
_CE_M=
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
MEMORY_PRESSURE_WRITE=c29tZSAyMDAwMDAgMjAwMDAwMAA=
XMODIFIERS=@im=ibus
DESKTOP_SESSION=ubuntu
GTK_MODULES=gail:atk-bridge
PWD=/home/rvu/shourrya_0100/operating_systems
```

Mar 31 14:51

rvu@rvu-OptiPlex-SFF-7020: ~/shourrya_0100/operating_systems

```
MEMORY_PRESSURE_WATCH=/sys/fs/cgroup/user.slice/user-1000.slice/user@1000.service/session.slice/org.gnome.Shell@wayland.service/memory.pressure
VTE_VERSION=7600
WAYLAND_DISPLAY=wayland-0
CONDA_PROMPT_MODIFIER=(base)
GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/1283f74c_7ca9_470b_86fd_c704fcf330c8
QTWEBENGINE_DICTIONARIES_PATH=/usr/share/hunspell-bdic/
GNOME_SETUP_DISPLAY=:1
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=xterm-256color
_CE_CONDA=
LESSOPEN=| /usr/bin/lesspipe %s
USER=rvu
GNOME_TERMINAL_SERVICE=:1.114
CONDA_SHLVL=1
DISPLAY=:0
SHLVL=2
GSM_SKIP_SSH_AGENT_WORKAROUND=true
QT_IM_MODULE=ibus
CONDA_PYTHON_EXE=/home/rvu/anaconda3/bin/python
XDG_RUNTIME_DIR=/run/user/1000
CONDA_DEFAULT_ENV=base
DEBUGINFOD_URLS=https://debuginfod.ubuntu.com
XDG_DATA_DIRS=/usr/share/ubuntu:/usr/share/gnome:/usr/local/share/:/usr/share/:/var/lib/snapd/desktop
PATH=/home/rvu/anaconda3/bin:/home/rvu/anaconda3/condabin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap
/bin:/snap/bin
GDMSESSION=ubuntu
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
OLDPWD=/home/rvu/shourrya_0100
_=/usr/bin/env
-----Environment Variables Manager------
1.Display Variables(printenv)
2. Set New Variables
3.Display variables(env)
4.Exit
Enter choice: 4
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$
```

Experiment 9:

```
Mar 31 14:54
rvu@rvu-OptiPlex-SFF-7020: ~/shourrya_0100/operating_systems
  GNU nano 7.2                                   IO.sh
#!/bin/bash

while true
do
        echo "-----I/O Redirection Manager-----"
        echo "1.Write to file "
        echo "2.append to file "
        echo "3.Display a file "
        echo "4.Exit"

read -p "Enter choice: " ch

case $ch in
1)
        read -p "Enter filename: " file
        echo "Enter text (CTRL+D to save): "
        cat > $file
        ;;

2)
        read -p "Enter filename: " file
        echo "Enter text to append (CTRL+D)"
        cat >> $file
        ;;

3)
        read -p "Enter filename: " file
        cat $file
        ;;

4)
        exit
        ;;

^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^Q Where Was
```

```
Mar 31 14:57
rvu@rvu-OptiPlex-SFF-7020: ~/shourrya_0100/operating_systems
 background_task.sh   cp.sh          IO.sh    lab3-2.sh        perm1.sh   perm.sh              process.sh
 child_1.sh           ev.sh          l3.sh    lu6999oyuj.tmp   perm2.sh   process_control.sh   'Screenshot from 2026-02-17 14-12-43-2.pdf'
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$ nano IO.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$ chmod +x IO.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$ ./IO.sh
-----I/O Redirection Manager-----
1.Write to file
2.append to file
3.Display a file
4.Exit
Enter choice: 1
Enter filename: hihi.txt
Enter text (CTRL+D to save):
hi this is a new file
-----I/O Redirection Manager-----
1.Write to file
2.append to file
3.Display a file
4.Exit
Enter choice: 2
Enter filename: hihi.txt
Enter text to append (CTRL+D)
hi hi-----I/O Redirection Manager-----
1.Write to file
2.append to file
3.Display a file
4.Exit
Enter choice: 3
Enter filename: hihi.txt
hi this is a new file
hi hi-----I/O Redirection Manager-----
1.Write to file
2.append to file
3.Display a file
4.Exit
Enter choice: 4
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$
```

Experiment 10:

```
Mar 31 14:58
rvu@rvu-OptiPlex-SFF-7020: ~/shourrya_0100/operating_systems
  GNU nano 7.2                                   cp.sh
#!/bin/bash

while true
 do
        echo "-----Command Piping Demo-----"
        echo "1.Count Files in directory"
        echo "2.search running processes"
        echo "3.show largest files"
        echo "4.exit"

read -p "Enter Choice " ch

case $ch in
        1)
                ls|wc -l
                ;;
        2)
                read -p "Enter process name: " pname
                ps -ef|grep $pname
                ;;
        3)
                du -ah|sort -rh|head -5
                ;;
        4)
                exit
                ;;
        *)
                echo "Invalid choice "
                ;;
        esac
done

                                        [ Read 31 lines ]
^G Help     ^O Write Out   ^W Where Is    ^K Cut       ^T Execute    ^C Location   M-U Undo     M-A Set Mark   M-] To Bracket
^X Exit     ^R Read File   ^\ Replace     ^U Paste     ^J Justify    ^/ Go To Line M-E Redo     M-6 Copy       ^Q Where Was
```

```
Mar 31 14:59
rvu@rvu-OptiPlex-SFF-7020: ~/shourrya_0100/operating_systems
 child_pro.sh        hi.txt     lab3-2.sh   perm2.sh        process_creation_1RUA25BCA0100.pdf
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$ nano cp.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$ chmod +x cp.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$ ./cp.sh
-----Command Piping Demo-----
1.Count Files in directory
2.search running processes
3.show largest files
4.exit
Enter Choice 1
22
-----Command Piping Demo-----
1.Count Files in directory
2.search running processes
3.show largest files
4.exit
Enter Choice 2
Enter process name: hih.txt
rvu         7320    7315  0 14:58 pts/0    00:00:00 grep hih.txt
-----Command Piping Demo-----
1.Count Files in directory
2.search running processes
3.show largest files
4.exit
Enter Choice 3
988K    .
452K    ./Screenshot from 2026-02-17 14-12-43-2.pdf
452K    ./process_creation_1RUA25BCA0100.pdf
4.0K    ./process.sh
4.0K    ./process_control.sh
-----Command Piping Demo-----
1.Count Files in directory
2.search running processes
3.show largest files
4.exit
Enter Choice 4
(base) rvu@rvu-OptiPlex-SFF-7020:~/shourrya_0100/operating_systems$
```